

If $z = y - x$, then the general solution of $y' = 1 + \frac{2(y-x)}{\ln(y-x)}$ is $x =$

Select one:

- ☐ a. $\frac{1}{4}(\ln z)^2 + c$
- ☐ b. $\frac{1}{2}(\ln z)^2 + c$
- ☐ c. $(\ln z)^2 + c$
- ☐ d. $\frac{1}{3}z^2 + c$

The general solution of
 $(x \cot(\frac{y}{x}) + y)dx - xdy = 0$ is

Select one:

- ☐ a. $x \sec(\frac{y}{x}) = c$
- ☐ b. $x \csc(\frac{y}{x}) = c$
- ☐ c. $x \cos(\frac{y}{x}) = c$
- ☐ d. $\cos(\frac{y}{x}) = cx$

The differential equation
 $y^2 y' + 3xy^3 = 3x^2 y^4$ is

Select one:

- ☐ a. Exact and Linear
- ☐ b. Separable and Nonlinear
- ☐ c. Nonlinear only
- ☐ d. Linear only

If the roots of the characteristic equation of a seventh order H.L.D.E. with constant coefficients are all repeated

$\lambda_1 = \lambda_2 = \dots = \lambda_7 = \lambda$, then the fifth solution will be of the form

Select one:

☐ a. $x^4 e^{\lambda x}$

☐ b. $x^5 e^{\lambda x}$

☐ c. $x^6 e^{\lambda x}$

☐ d. $x^7 e^{\lambda x}$

Suppose that $u = \frac{y}{x}$, then the general solution of $y' = \frac{y}{x} + \left(\frac{y}{x}\right)^3$ is

Select one:

- ☐ a. $\ln(x) = \frac{u^4}{4} + c$
- ☐ b. $\ln(x) = \frac{-1}{2u^2} + c$
- ☐ c. $\ln(x) = \frac{1}{u^2} + c$
- ☐ d. $\ln(x) = \frac{-u^2}{2} + c$

Which of the following functions is a solution to $xy' + y = 3x^2$

Select one:

- ☐ a. e^{-x}
- ☐ b. x^3
- ☐ c. x^2
- ☐ d. x



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The integrating factor of
 $2xy' + 2xy + y = 3e^x$ is $\mu =$

Select one:

- ☐ a. $x^{-1}e^x$
- ☐ b. xe^x
- ☐ c. $\sqrt{x}e^x$
- ☐ d. $\sqrt{x}e^{2x}$

If $M(x, y)dx + N(x, y)dy = 0$ is both exact and homogeneous, then the general solution is

Select one:

- ☐ a. $xM - yN = c$
- ☐ b. $yM + xN = c$
- ☐ c. $yM - xN = c$
- ☐ d. $xM + yN = c$

The general solution of $y''' - 4y'' + 5y' - 2y = 0$ is $y =$

Select one:

- ☐ a. $c_1 e^x + c_2 x e^x + c_3 e^{-2x}$
- ☐ b. $c_1 e^{-x} + c_2 x e^{-x} + c_3 e^{2x}$
- ☐ c. $c_1 x e^{-x} + c_2 x^2 e^x + c_3 e^{2x}$
- ☐ d.

$$c_1 e^x + c_2 x e^x + c_3 e^{2x}$$

The differential equation $xydy + (x^3y^3 - 2y^2)dx = 0$ is

Select one:

- ☐ a. Homogeneous
- ☐ b. Exact
- ☐ c. Linear
- ☐ d. Bernoulli

The general solution of $y' + \frac{5}{x}y = 2$ is $y =$

Select one:

- ☐ a. $\frac{1}{3x} + \frac{c}{x^5}$
- ☐ b. $\frac{x}{3}$
- ☐ c. $\frac{x}{3} + \frac{c}{x^5}$
- ☐ d. $\frac{x}{6} + \frac{c}{x^4}$



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The general solution of the exact differential equation

$$(2ye^{2x} - y\cos(xy))dx + (e^{2x} - x\cos(xy))dy = 0$$

is

Select one:

- ☐ a. $ye^{2x} + \sin(xy) = c$
- ☐ b. $ye^{2x} - \sin(xy) = c$
- ☐ c. $ye^{2x} + \cos(xy) = c$
- ☐ d. $xe^{2y} - \sin(xy) = c$

If $y = c_1x^2 + c_2$ is the general solution to a second order D.E. Then the particular solution of the same D.E. such that $y(1) = 3, y'(1) = 2$ is $y =$

Select one:

- ☐ a. x^2
- ☐ b. $2x^2 + 2$
- ☐ c. $x^2 + 1$
- ☐ d. $x^2 + 2$

If the roots of the characteristic equation of a H.L.D.E. with constant coefficients are $1, -2$, then the D.E. is

Select one:

☐ a. $y'' + y' + 2y = 0$

☐ b. $y'' - y' + 2y = 0$

☐ c. $y'' - y' - 2y = 0$

☐ d. $y'' + y' - 2y = 0$

The D.Equation $M(x, y)dx + N(x, y)dy = 0$ is exact if $M_x = N_y$

Select one:

☐ True

☒ False



The equation $\frac{dy}{dx} = \frac{x^2 + xy + y^2}{x^2}$ is homogeneous

Select one:

- ☒ True
- ☐ False



The substitution $z = y^{-3}$ transforms the equation $\frac{dy}{dx} = xy^4 - 1$ into separable D.Equation

Select one:

- ☒ True
- ☐ False



The general solution of $\frac{dy}{dx} = \frac{x+y+3}{2x+2y+6}$ is

Select one:

☐ a. $y = x^2 + c$

☐ b. $y = x^3 + c$

☐ c. $y = 2x + c$

☒ d. $y = \frac{1}{2}x + c$

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The integrating factor for $ydx + (2x - ye^y)dy = 0$ is

Select one:

- ☒ a. y
- ☐ b. x
- ☐ c. $-x$
- ☐ d. $-y$



The equation

$$(y \cos x + 2xe^y)dx + (\sin x + x^2 e^y - 1)dy = 0$$

is

Select one:

- ☐ a. Bernoulli
- ☒ b. Exact
- ☐ c. None
- ☐ d. linear



The D.Equation $M(x, y)dx + N(x, y)dy = 0$ is exact if $M_x = N_y$

Select one:

- ☐ True
- ☒ False



The general solution of $\frac{dy}{dx} = \frac{x+y+3}{2x+2y+6}$ is

Select one:

- ☐ a. $y = x^2 + c$
- ☐ b. $y = x^3 + c$
- ☐ c. $y = 2x + c$
- ☒ d. $y = \frac{1}{2}x + c$

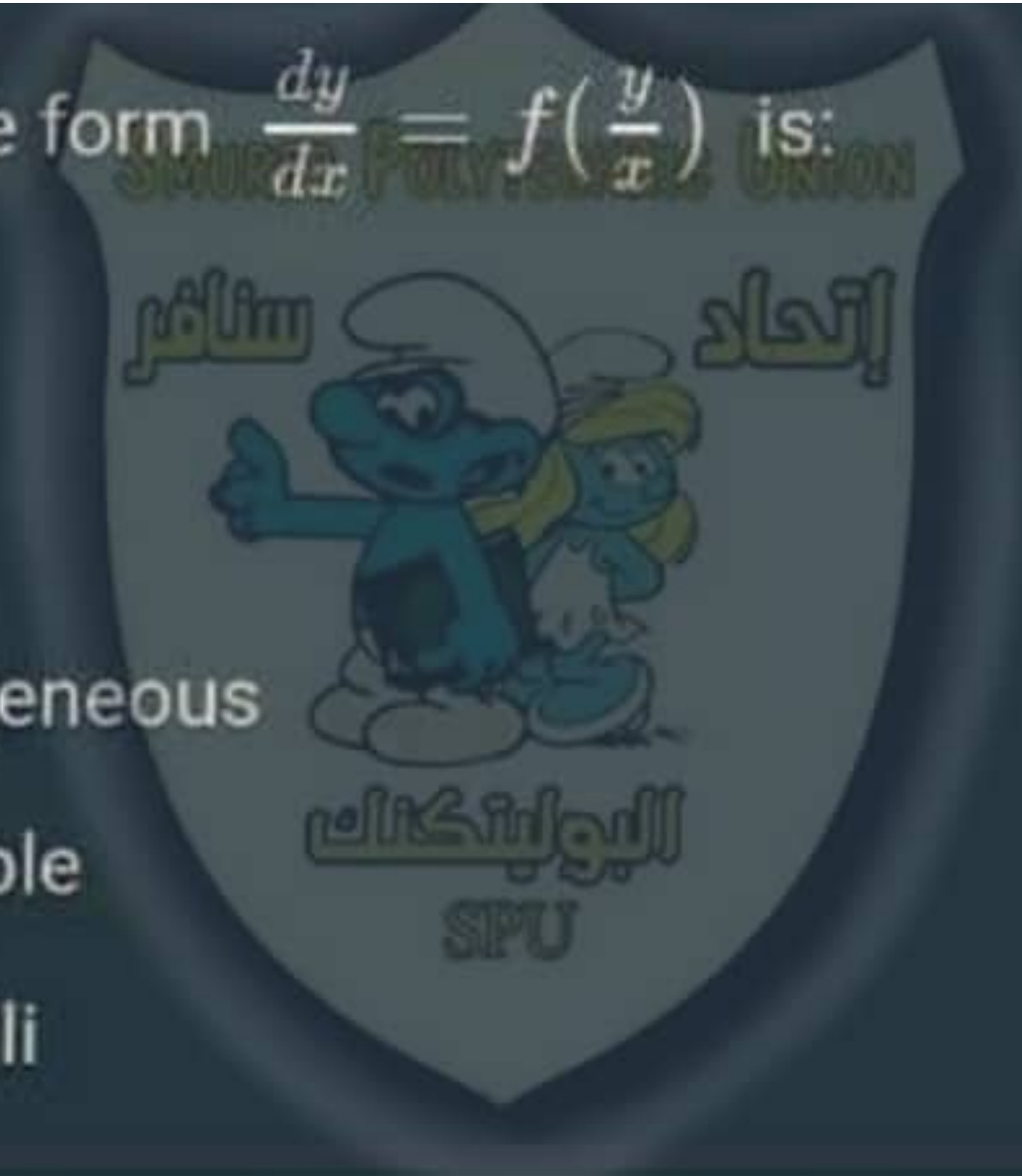
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The ODE in the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ is:

Select one:

- ☐ a. Linear
- ☒ b. Homogeneous
- ☐ c. Separable
- ☐ d. Bernoulli



The solution of $\frac{dy}{dx} + 2y = e^x, y(0) = 0$ is

Select one:

☐ a. $\frac{1}{3}(e^{2x} - e^x)$

☐ b. $\frac{1}{3}(e^x - e^{2x})$

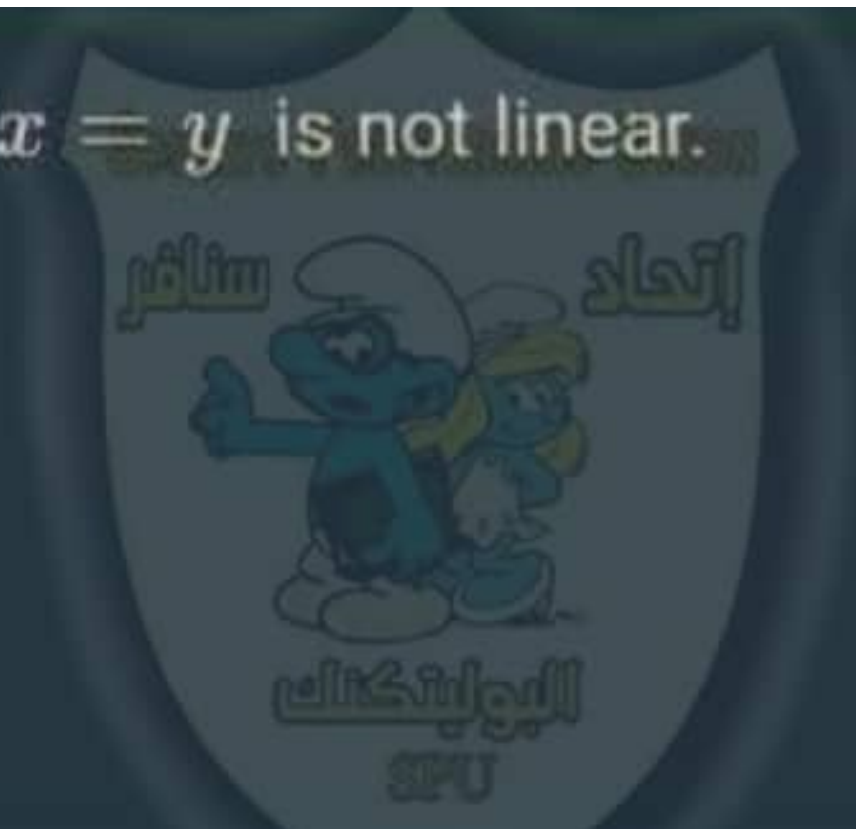
☐ c. $\frac{1}{3}(e^{2x} - e^{-2x})$

☒ d. $\frac{1}{3}(e^x - e^{-2x})$

The ODE $\frac{dy}{dx} + x = y$ is not linear.

Select one:

- ☐ True
- ☒ False



The integrating factor for the linear equation

$$y' + (\cot x)y = 1 \text{ is } |\sin x|$$

Select one:

- ☒ True
- ☐ False



The integrating factor that converts the ODE $(3x^2y + 2xy + y^3)dx + (x^2 + y^2)dy = 0$ into exact is $\mu(x) =$

Select one:

- ☐ a. e^x
- ☐ b. e^{6x}
- ☒ c. e^{3x}
- ☐ d. e^{2x}



The integrating factor transforms the non exact D.
Equ. into

Select one:

- ☐ a. linear
- ☐ b. homogeneous
- ☒ c. exact
- ☐ d. Bernoulli



If $M(x, y)dx + (2xy + y^2)dy = 0$ is an exact ODE, then $M(x, y) =$

Select one:

☐ a. x^2

☒ b. y^2

☐ c. y

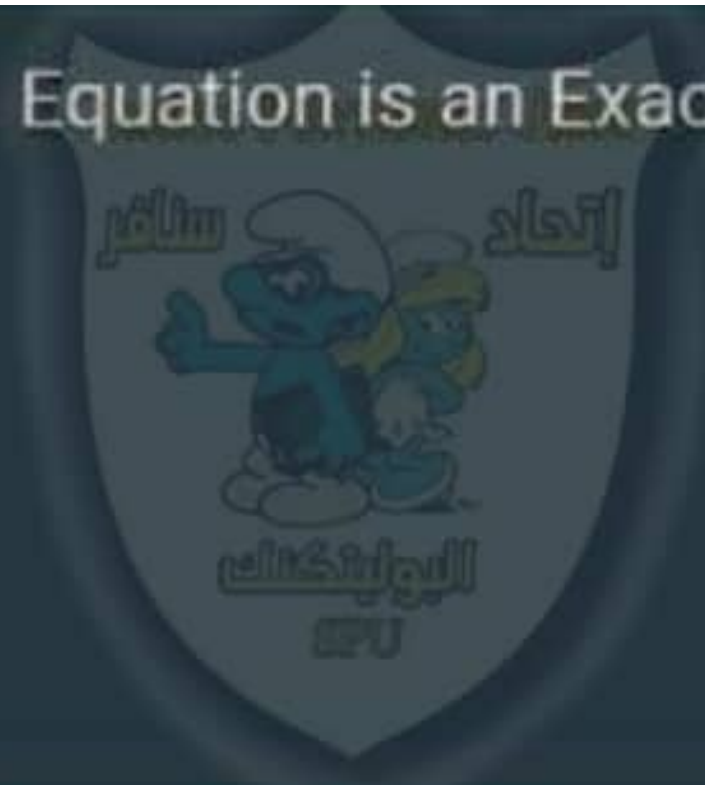
☐ d. $2x$



Every separable D. Equation is an Exact D. Equation

Select one:

- ☐ True
- ☒ False



The ODE $\frac{dy}{dx} + \frac{y}{x} = \sqrt{y}$ is Bernoulli's equation.

Select one:

- ☒ True
- ☐ False



If $\frac{M_y - N_x}{N} = f(x)$. Then the integrating factor is

Select one:

- ☐ a. $e^{\int 4f(x)dx}$
- ☒ b. $e^{\int f(x)dx}$
- ☐ c. $2x$
- ☐ d. $e^{\int -f(x)dx}$

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The ODE $x^2 y dx + (\frac{1}{3} x^3 + \sin y) dy = 0$ is

Select one:

- ☐ a. Homogeneous
- ☒ b. Exact
- ☐ c. Separable
- ☐ d. Not exact

The Bernoulli D. Equation is an extension to the separable D. Equation

Select one:

- ☐ True
- ☒ False

Time left 0:09:50

Question 1

Not yet answered

Marked out of 2.00

Flag question

If $y'''' - y = \cos x + 2\sin x$, then the particular solution $y_p =$

- ☐ a. $e^x(c_1\cos x + c_2\sin x)$
- ☐ b. $x^2(c_1\cos x + c_2\sin x)$
- ☐ c. $xe^x(c_1\cos x + c_2\sin x)$
- ☒ d. $x(c_1\cos x + c_2\sin x)$
- ☐ e. $c_1\cos x + c_2\sin x$

[Clear my choice](#)

If the general solution of

$$x^2 y'' - 3xy' + 3y = 2x^4 e^x \text{ is } y = y_h + y_p,$$

then the particular solution (by variation of parameters) $y_p =$

- ☐ a. $(2x - 2)e^x$
- ☐ b. $(x^3 + 2x^2 - 2x)e^x$
- ☐ c. $(2x^2 + 2x)e^x$
- ☐ d. $(x^2 - 2x + 3)e^{-x}$
- ☒ e. $(2x^2 - 2x)e^x$

[Clear my choice](#)

Question **3**

Not yet answered

Marked out of 2.00

🚩 Flag question

If $\frac{dy}{dx} = 4x^3 y^{-2}$, $y(0) = 0$, then $y(\sqrt{3}) =$

- ☐ a. $\sqrt[3]{3}$
- ☐ b. $\sqrt{3}$
- ☒ c. 3
- ☐ d. -3
- ☐ e. 2

[Clear my choice](#)

Question 1

Not yet answered

Marked out of 2.00

🚩 Flag question

The general solution of the ODE $\frac{dy}{dx} = \frac{x^2 e^{\frac{-2y}{x}} + y^2}{xy}$ is:

- ☐ a. $(\frac{y}{2x} + 2x)e^{\frac{2y}{x}} = \ln|x| + c$
- ☐ b. $(\frac{y}{2x} + \frac{y}{4})e^{\frac{2y}{x}} = \ln|x| + c$
- ☐ c. $(\frac{y}{2x} - \frac{x}{4})e^{\frac{2y}{x}} = \ln|x| + c$
- ☐ d. $(\frac{y}{2x} + \frac{x}{2})e^{\frac{2y}{x}} = \ln|x| + c$
- ☒ e. $(\frac{y}{2x} - \frac{1}{4})e^{\frac{2y}{x}} = \ln|x| + c$

Clear my choice

Question 2

Not yet answered

Marked out of 2.00

🚩 Flag question

If $\frac{dy}{dx} = \sqrt{x+y} - 1$, $y(0) = 1$, then $y(-2) =$

- ☐ a. 3
- ☐ b. 5
- ☐ c. 4
- ☐ d. 9
- ☒ e. 2

[Clear my choice](#)

Given that :

$$\frac{dy}{dx} = 2x + 2xy^2, \quad y(0) = 0$$

then

$$y\left(\sqrt{\frac{3}{4}}\pi\right) =$$

Select one:

- ☐ a. 1
- ☐ b. $\sqrt{3}$
- ☒ c. -1
- ☐ d. $-\sqrt{3}$

[Clear my choice](#)

The assumption, $u = \frac{y}{x}$ converts the differential equation ;

$$\frac{dy}{dx} = \frac{x+y}{x-y} \text{ into}$$

Select one:

☐ a. $x \frac{du}{dx} = \frac{1-u^2}{2-u}$

☐ b. $x \frac{du}{dx} = \frac{1+u^2}{2-u}$

☐ c. $x \frac{du}{dx} = \frac{2+u^2}{1-u}$

☒ d. $x \frac{du}{dx} = \frac{1+u^2}{1-u}$

[Clear my choice](#)

If the order and the degree of the differential equation;

$$(y''')^4 + (y'')^5 + (y')^6 = e^x$$

are a and b respectively, then the order pair $(a, b) =$

Select one:

- ☒ a. $(3, 4)$
- ☐ b. $(6, 3)$
- ☐ c. $(3, 6)$
- ☐ d. $(4, 3)$

[Clear my choice](#)

The integrating factor that makes the differential equation;

$$2xdx + \left(\frac{x^2}{\sqrt{1-y^2}} + y \right) dy = 0$$

to be exact, is:

Select one:

- ☒ a. $e^{\sin^{-1} y}$
- ☐ b. $\tan^{-1} y$
- ☐ c. $\sin^{-1} y$
- ☐ d. $e^{\tan^{-1} y}$

[Clear my choice](#)

The general solution of the differential equation;

$$x \frac{dy}{dx} + 2y = 4x\sqrt{y} \quad , x > 0$$

is:

Select one:

- ☐ a. $(x + \frac{c}{x})$, where c is an arbitrary constant.
- ☐ b. $(2x + \frac{c}{x})^2$, where c is an arbitrary constant.
- ☒ c. $(x + \frac{c}{x})^2$, where c is an arbitrary constant.
- ☐ d. $(2x + \frac{c}{x})$, where c is an arbitrary constant.

[Clear my choice](#)